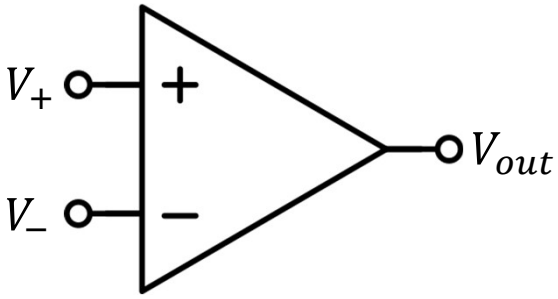
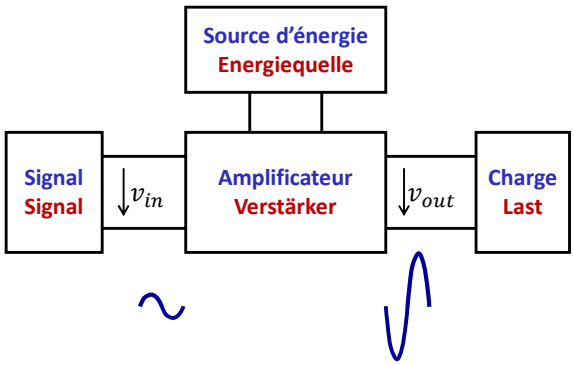
Amplificateur Opérationnel (Ampli OP)
Operationsverstärker (Op Amp)
Operational Amplifier (Op Amp)

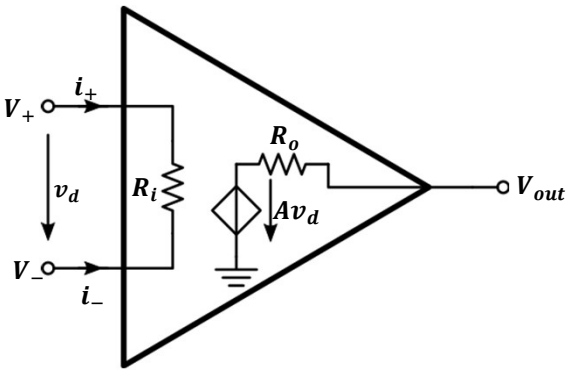


Principe d'amplification des signaux:
Prinzip der Signalverstärkung:



Amplificateur Opérationnel idéal
Idealer Operationsverstärker

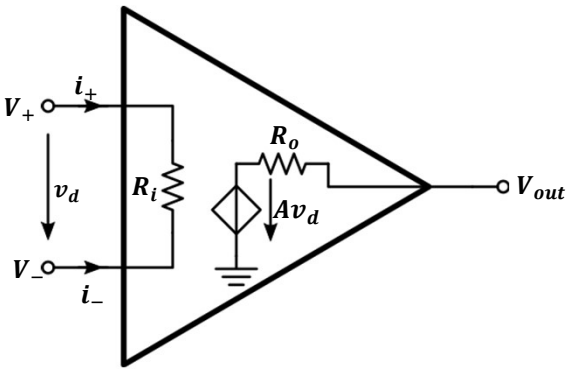
$$A = \infty$$
$$R_0 = 0$$
$$R_i = \infty$$
$$i_1 = i_2 = 0$$



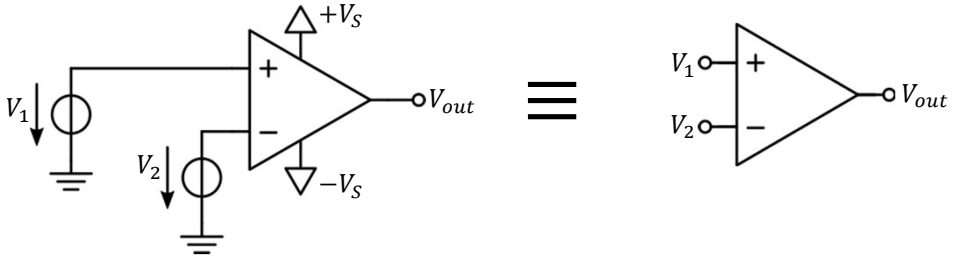
Amplificateur opérationnel réel
Realer Operationsverstärker

Exemple
Beispiel
LM741

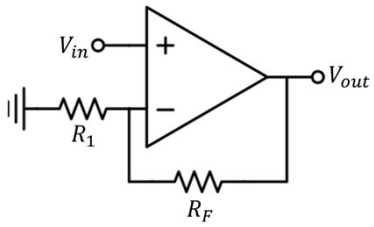
$$A = 10^5$$
$$R_0 = 75\Omega$$
$$R_i = 2\ M\Omega$$
$$i_1 = i_2 \neq 0$$



Circuit de l'amplificateur opérationnel
Schaltung eines Operationsverstäkers



Exercice 1a
Aufgabe 1a



$V_{out} = A(V_+ - V_-)$

$A = 10^5$

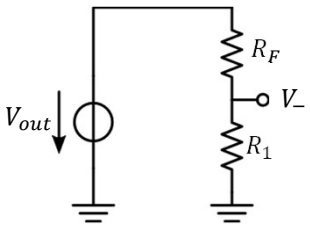
$i_+ = i_- = 0$

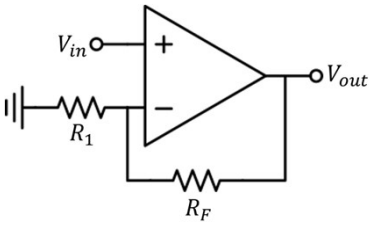
$V_+ =$

$V_- =$

$V_{out} = A(V_+ - V_-) \rightarrow$

$V_{out} =$





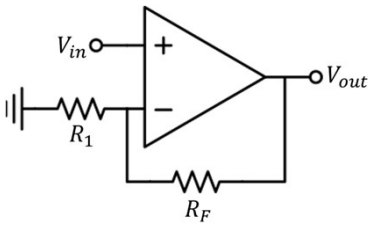
$$V_{out} = A(V_+ - V_-)$$

$$A = 10^5$$

$\frac{V_{out}}{V_{in}} = \frac{R_1 + R_F}{R_1}$	$\frac{V_{out}}{V_{in}} = \frac{A}{1 + A\frac{R_1}{R_1 + R_F}}$
1	0.99999
10	9.99900
100	99.9001
1000	990.099
10000	9090.91
100000	50000

Exercice 1b

Aufgabe 1b



$$V_{out} = A(V_+ - V_-)$$

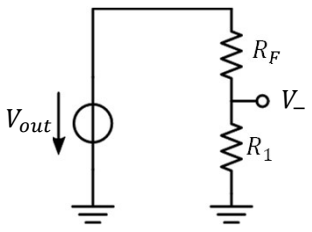
$$A \to \infty \Rightarrow V_+ \to V_-$$

$$\boxed{V_+ = V_-}$$

$V_+ =$

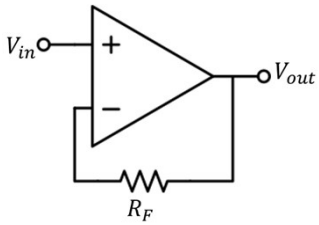
$V_- =$

$V_+ = V_-$ ⇨



$V_{out} =$

Exercice 1c
Aufgabe 1c



$V_{out} = A(V_+ - V_-)$

$A \to \infty$ ➡ $V_+ \to V_-$

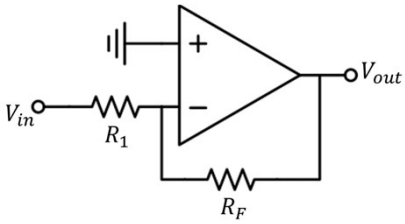
$\boxed{V_+ = V_-}$

$V_+ =$

$V_- =$

$V_+ = V_-$ ➡

Exercice 2
Aufgabe 2



$V_{out} = A(V_+ - V_-)$

$A \to \infty$ ➡ $V_+ \to V_-$

$\boxed{V_+ = V_-}$

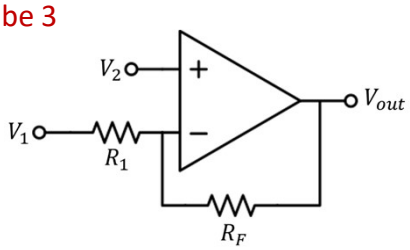
$V_+ = 0$

$V_- =$

$V_+ = V_-$ ➡

$v_{out} =$

Exercice 3
Aufgabe 3



$$V_+ = V_-$$

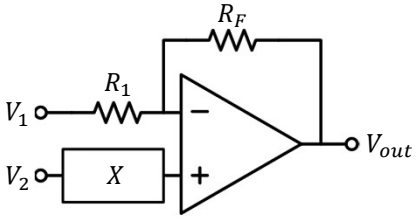
$V_+ =$

$V_- =$

$V_+ = V_-$ ⇒

$V_{out} =$

Exercice 4
Aufgabe 4



$$V_+ = V_-$$

$$V_{out} = (V_2 - V_1)\frac{R_F}{R_1}$$

$V_+ = X \times V_2$

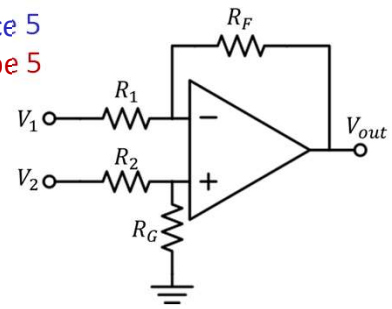
$V_- =$

$V_+ = V_-$

$V_{out} =$

$X =$

Exercice 5
Aufgabe 5



$V_+ =$ $V_- =$

$V_+ = V_-$

$V_{out} =$

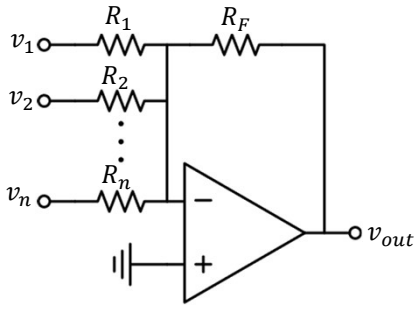
$R_G =$

$V_+ = V_-$ $R_G = ?$

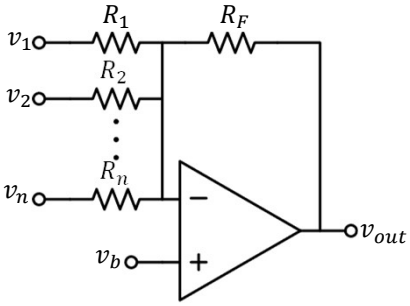
$$V_{out} = V_2 \frac{R_F}{R_2} - V_1 \frac{R_F}{R_1}$$

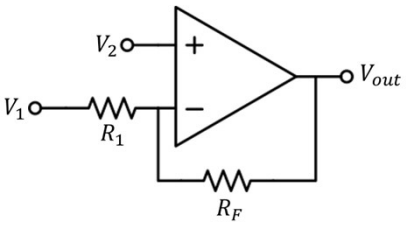
$$V_{out} = V_2 G_2 - V_1 G_1$$

Inverseur sommateur
Invertierender Addierer



Inverseur sommateur général
Allgemeiner Invertierender Addierer





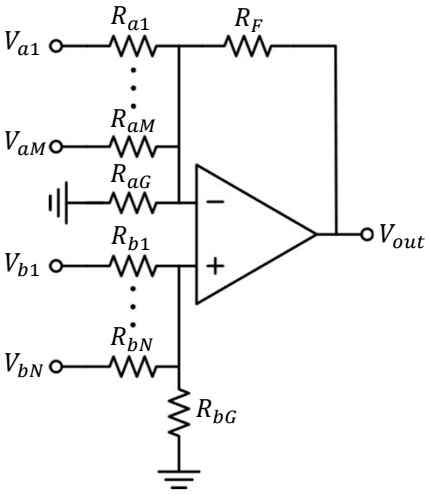
$$V_{out} = V_2 \frac{R_1 + R_F}{R_1} - V_1 \frac{R_F}{R_1}$$

$$V_{out} = -\left(V_1 \frac{R_F}{R_1} + \cdots + V_n \frac{R_F}{R_n}\right) + ???$$

$$\boxed{V_{out} = -\left(V_1 \frac{R_F}{R_1} + \cdots + V_n \frac{R_F}{R_n}\right) + V_b \left(\frac{R_F}{R_1 \parallel \cdots \parallel R_n} + 1\right)}$$

$$\boxed{V_{out} = -\left(V_1 \frac{R_F}{R_1} + \cdots + V_n \frac{R_F}{R_n}\right) + V_b \left(\frac{R_F}{R_1} + \cdots + \frac{R_F}{R_n} + 1\right)}$$

Additionneur soustracteur general
Allgemeiner Addierer-Subtrahierer



$$G_{bi} = \frac{R_F}{R_{bi}} \qquad G_{ai} = \frac{R_F}{R_{ai}}$$

$$V_{out} = \sum_{i=1}^{N} G_{bi} V_{bi} - \sum_{i=1}^{M} G_{ai} V_{ai}$$

Si / wenn

$$R_{aG} = \infty \qquad R_{bG} = R_F \frac{1}{1 + \sum_{i=1}^{M} G_{ai} - \sum_{i=1}^{N} G_{bi}}$$

ou / oder

$$R_{bG} = \infty \qquad R_{aG} = R_F \frac{1}{\sum_{i=1}^{N} G_{bi} - \sum_{i=1}^{M} G_{ai} - 1}$$